Xin Xiong

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CV Updated: October 2025

Education

- **PhD in Computational Biology**
  Hong Kong Baptist University, September 2025
  Advisor: Professor Liang Tian
  Thesis: Deconvolution and Reconstruction of the Tumor Microenvironment Using Deep Learning and scRNA-seq Data

- **Master of Engineering in Computer Technology**
  Shanghai Jiao Tong University, 2021
  Advisor: Professor Hai Zhao
  Thesis: Word Vector-Based Representation and Classification of Small Molecule Compounds

- **Bachelor of Engineering in Bioengineering**
  Xi'an Polytechnic University, 2010

Work Experience

- **Post-Doctoral Research Fellow**
  Hong Kong Baptist University, October 2025 – Present
- **Visiting Student (Prof. Nicolas Chevrier's Group)**
  University of Chicago, April 2025 – June 2025
  - Analyzed organ interactions in colitis models using multi-tissue transcriptomic data.
- **Research Assistant**
  Shenzhen Institutes of Advanced Technology (SIAT), CAS, 2020 – 2021
  - Developed deconvolution methods for bulk RNA-seq data from primary solid tumors.
  - Investigated relationships between gene expression programs (e.g., EMT) and immune cell infiltration in the tumor microenvironment.
- **Research Assistant**
  Southern University of Science and Technology (SUSTech), 2019 – 2020
  - Contributed to a project on single-guide RNA (sgRNA) design for CRISPR/Cas9-mediated gene editing.
- **Research Assistant**
  Interdisciplinary Research Center of Biology and Chemistry (IRCBC), CAS, 2016 – 2019
  - Developed machine learning algorithms for predicting collision cross-section (CCS) values in mass spectrometry.
  - Contributed to the development of web servers for metabolomics and lipidomics (MetCCS and LipidCCS).

- **Bioinformatician**
  Genminix Informatics Ltd.Co., 2011 – 2016
  - Developed bioinformatic workflows and algorithms for RNA-seq data analysis.
  - Invented a patented computational method for rapid identification and visualization of feed-forward loops in gene regulatory networks (Chinese Patent No. 201410112193X).

Teaching Experience

- **Teaching Assistant**, Thermal and Statistical Physics (PHYS3047), 2023 – 2024
- **Teaching Assistant**, Thermal and Statistical Physics (PHYS3047), 2022 – 2023

Honors

- 2024 Best Speaker Award, Peking University Shenzhen Graduate School, China
- 2010 Outstanding Graduation Thesis, Xi'an Polytechnic University, China
- 2010 Outstanding Class Leader, Xi'an Polytechnic University, China
- 2008 – 2010 Encouragement Scholarship, Xi'an Polytechnic University, China

Presentations

- **DeSide2: A Deep Learning Framework for Precise Gene Expression Deconvolution in the Tumor Microenvironment**.
  Oral Presentation, *29th International Conference on Statistical Physics (StatPhys29)*, July 13–18, 2025, Florence, Italy.
- **DeSide2: A Deep Learning Framework for Precise Gene Expression Deconvolution in the Tumor Microenvironment**.
  Oral Presentation, *HKBU International Advisory Board of Graduate Studies Annual Meeting*, June 25–26, 2025, Hong Kong Baptist University, Hong Kong, China.
- **DeSide: A Unified Deep Learning Approach for Cellular Deconvolution of the Tumor Microenvironment**.
  Oral Presentation, *1st AI for Science PhD Student Academic Forum in the Guangdong-Hong Kong-Macao Greater Bay Area*, November 1, 2024, Peking University Shenzhen Graduate School, Shenzhen, China.
- **DeSide: A Unified Deep Learning Approach for Cellular Decomposition of Bulk Tumors Based on Limited scRNA-seq Data**.
  Poster Presentation, *28th International Conference on Statistical Physics (StatPhys28)*, August 7–11, 2023, University of Tokyo, Tokyo, Japan.

Publications

$ denotes equal contributions; * denotes corresponding author(s)

- **Xiong, X.**[$], Liu, Y.[$], Pu, D., Yang, Z., Bi, Z., Tian, L.[*], and Li, X.[*] (2024). DeSide: A unified deep learning approach for cellular deconvolution of tumor microenvironment. *Proc. Natl. Acad. Sci. U. S. A.* 121, e2407096121. GitHub
- Zhou, Z., Luo, M., Chen, X., Yin, Y., **Xiong, X.**, Wang, R., and Zhu, Z.-J.[*] (2020). Ion mobility collision cross-section atlas for known and unknown metabolite annotation in untargeted metabolomics. *Nat. Commun.* 11, 4334.
- Shen, X., Wang, R., **Xiong, X.**, Yin, Y., Cai, Y., Ma, Z., Liu, N., and Zhu, Z.-J.[*] (2019). Metabolic reaction network-based recursive metabolite annotation for untargeted metabolomics. *Nat. Commun.* 10, 1516.
- Zhou, Z., Shen, X., Chen, X., Tu, J., **Xiong, X.**, and Zhu, Z.J.[*] (2019). LipidIMMS Analyzer: integrating multi-dimensional information to support lipid identification in ion mobility—mass spectrometry based lipidomics. *Bioinformatics* 35.
- Zhou, Z., Tu, J., **Xiong, X.**, Shen, X., and Zhu, Z.J.[*] (2017). LipidCCS: prediction of collision cross-section values for lipids with high precision to support ion mobility–mass spectrometry-based lipidomics. *Anal. Chem.* 89, 9559–9566. Website
- Zhou, Z., **Xiong, X.**, and Zhu, Z.-J.[*] (2017). MetCCS predictor: a web server for predicting collision cross-section values of metabolites in ion mobility-mass spectrometry based metabolomics. *Bioinformatics* 33, 2235–2237. Website

References

- **Prof. Liang Tian**
  Associate Professor
  Department of Physics
  Hong Kong Baptist University
  Email: liangtian@hkbu.edu.hk
  Relationship: Principal Supervisor, Ph.D. studies

- **Prof. Xuefei Li**
  Principal Investigator
  Shenzhen Institutes of Advanced Technology (SIAT)
  Chinese Academy of Sciences
  Email: xuefei.li@siat.ac.cn
  Relationship: Co-supervisor, DeSide Project

- **Dr. Qianyuan Tang**
  Assistant Professor
  Department of Physics
  Hong Kong Baptist University
  Email: tangqy@hkbu.edu.hk
  Relationship: Ph.D. Candidature Defense Committee Member

- **Prof. Nicolas Chevrier**
  Associate Professor
  Pritzker School of Molecular Engineering
  The University of Chicago
  Email: nchevrier@uchicago.edu
  Relationship: Mentor during visiting study